Appendix R Script

```
library(tidyverse)
library(rio)
library(haven)
library(knitr)
library(DT)
library(huxtable)
library(lmtest)
library(sandwich)
library(here)
library(dplyr)
library(broom)
library(gridExtra)
library(ggridges)
library(ggplot2)
```

```
# Load data civ88-08_nutri.dta and civ88-08_mortality.dta
civ_nutri <- read_dta(here::here("measurement_p2/data/civ88-08_nutri.dta"))
civ_mortality <- read_dta(here::here("measurement_p2/data/civ88_mort.dta"))
# save as csv
# write_csv(civ_nutri, "measurement_p2/data/civ88-08_nutri.csv")
# write_csv(civ_mortality, "measurement_p2/data/civ88-08_mortality.csv")
```

Exercise 1

1.1.a

```
# Method 1
# Boys
model_1_boys <- lm(
  (dboys_l5) / (boys_l5) ~ 1,
  data = civ_mortality %>% filter(boys_l5 > 0),
  weights = (pweight * n_wom) * boys_l5
)

# Girls
model_1_girls <- lm(
  (dgirls_l5) / (girls_l5) ~ 1,
  data = civ_mortality %>% filter(girls_l5 > 0),
  weights = (pweight * n_wom) * girls_l5
)

# Method 2
# Boys
model_2_boys <- lm(
  (dboys_u5 - dboys_l5) / (boys - boys_l5) ~ 1,
  data = civ_mortality %>% filter((boys - boys_l5) > 0),
  weights = (pweight * n_wom) * (boys - boys_l5)
)
```

```r
# Girls
model_2_girls <- lm(
  (dgirls_u5 - dgirls_l5) / (girls - girls_l5) ~ 1,
  data = civ_mortality %>% filter((girls - girls_l5) > 0),
  weights = (pweight * n_wom) * (girls - girls_l5)
)

# 1. Create Pooled Data for Boys
boys_pooled <- bind_rows(
  civ_mortality %>%
    filter(boys_l5 > 0) %>%
    transmute(rate = dboys_l5 / boys_l5, w = (pweight * n_wom) * boys_l5, is_m2 = 0),
  civ_mortality %>%
    filter((boys - boys_l5) > 0) %>%
    transmute(rate = (dboys_u5 - dboys_l5) / (boys - boys_l5), w = (pweight * n_wom) * (boys - boys_l5
)

# 2. Create Pooled Data for Girls
girls_pooled <- bind_rows(
  civ_mortality %>%
    filter(girls_l5 > 0) %>%
    transmute(rate = dgirls_l5 / girls_l5, w = (pweight * n_wom) * girls_l5, is_m2 = 0),
  civ_mortality %>%
    filter((girls - girls_l5) > 0) %>%
    transmute(rate = (dgirls_u5 - dgirls_l5) / (girls - girls_l5), w = (pweight * n_wom) * (girls - gi
)

# 3. Run the Difference Regressions
diff_boys <- lm(rate ~ is_m2, data = boys_pooled, weights = w)
diff_girls <- lm(rate ~ is_m2, data = girls_pooled, weights = w)

# 4. Extract the coefficient matrices
ct_boys <- coeftest(diff_boys)
ct_girls <- coeftest(diff_girls)

# 5. Isolate ONLY the 2nd row (the difference), and restore the 'coeftest' class
# so that broom/huxtable knows how to read it without throwing a matrix error.
ct_diff_boys <- ct_boys[2, , drop = FALSE]
class(ct_diff_boys) <- "coeftest"
rownames(ct_diff_boys) <- "(Intercept)"  # Disguise it for alignment

ct_diff_girls <- ct_girls[2, , drop = FALSE]
class(ct_diff_girls) <- "coeftest"
rownames(ct_diff_girls) <- "(Intercept)"

# 1. Create Pooled Data for Sex Comparison (Method 2 Only)
m2_sex_pooled <- bind_rows(
  # Boys Method 2 (Indicator = 0)
  civ_mortality %>%
    filter((boys - boys_l5) > 0) %>%
    transmute(rate = (dboys_u5 - dboys_l5) / (boys - boys_l5), w = pweight * n_wom *(boys - boys_l5),
  # Girls Method 2 (Indicator = 1)
  civ_mortality %>%
```

```
    filter((girls - girls_l5) > 0) %>%
    transmute(rate = (dgirls_u5 - dgirls_l5) / (girls - girls_l5), w = pweight * n_wom *(girls - girls
)

# 2. Run the Difference Regression (Girls M2 - Boys M2)
diff_sex_m2 <- lm(rate ~ is_girl, data = m2_sex_pooled, weights = w)

# 3. Isolate the gender difference coefficient and disguise it for huxreg alignment
ct_diff_sex_m2 <- coeftest(diff_sex_m2)[2, , drop = FALSE]
class(ct_diff_sex_m2) <- "coeftest"
rownames(ct_diff_sex_m2) <- "(Intercept)"

# 6. Generate the single-row 6-Column Table
outreg <- huxreg(
  `M1 (Boys)` = model_1_boys,
  `M2 (Boys)` = model_2_boys,
  `Diff (Boys)` = ct_diff_boys,
  `M1 (Girls)` = model_1_girls,
  `M2 (Girls)` = model_2_girls,
  `Diff (Girls)` = ct_diff_girls,
  `Diff M2 G-B` = ct_diff_sex_m2, # New column added here
  coefs = c("Mortality rate" = "(Intercept)"),
  error_format = "({std.error})",
  error_pos = "below",
  align = "c",
  statistics = c(N = "nobs", R2 = "r.squared"),
  stars = c(`***` = 0.01, `**` = 0.05, `*` = 0.1),
  note = "Significance: {stars}"
)

outreg <- set_label(outreg, "tbl-child_mortality")
outreg <- set_caption(outreg, "Child mortality rates in Côte d'Ivoire, by sex and method.")
outreg <- set_all_padding(outreg, 1.5)
outreg
```

Exercise 2

2.1.b

```
plot_1 <-ggplot(civ_nutri, aes(weightkg, fill = factor(year), colour = factor(year))) + ylab("Kernel D
  geom_density(alpha = 0.3) +
  labs(fill = "Year", colour = "Year")+
  xlim(0, 100) + ylim(0, NA) + theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle = element_text(hjust = 0.5), legend.posi

# update original data
civ_nutri <- civ_nutri %>% mutate(weightkg_corrected = ifelse(year == 2008 & weightkg > 25, weightkg /

plot_2 <-ggplot(civ_nutri, aes(weightkg_corrected, fill = factor(year), colour = factor(year))) + ylab
  geom_density(alpha = 0.3) +
  labs(fill = "Year", colour = "Year")+
```

Table 1

```r
# estimate weight mean by year
weight_mean_by_year <- civ_nutri %>%
  group_by(year) %>%
  # weighted.mean() does not remove missing survey weights. In 1988,
  # missing pweight values therefore make the result NA.
  summarise(
    observations = sum(!is.na(weightkg) & !is.na(pweight)),
    mean_weight = weighted.mean(
      weightkg[!is.na(weightkg) & !is.na(pweight)],
      pweight[!is.na(weightkg) & !is.na(pweight)]
    ),
    .groups = "drop"
  )

# now use the estimator from above that corrects for the hectogram issue in 2008
p_h <- 0.304 # assume 10% of weights are in hectograms
weight_mean_by_year <- weight_mean_by_year %>%
  mutate(mean_weight_corrected = ifelse(year == 2008, mean_weight / (1 + 9 * p_h), mean_weight))

# what percentage of observations in 2008 are above 25kg?
p_above_25 <- civ_nutri %>%
  filter(year == 2008) %>%
  summarise(percentage = mean(weightkg > 25, na.rm = TRUE) * 100)

# output the results in a nice table
kable(
  weight_mean_by_year,
  digits = 2,
  col.names = c("Year", "Obs", "Mean Weight", "Mean Weight Corrected"),
  caption = "Summary of Weight Estimates With Hectogram Correction"
)
```

```
  xlim(0, 100) + ylim(0, NA) + theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle = element_text(hjust = 0.5), legend.posi

grid.arrange(plot_1, plot_2, ncol=2)
```

2.2

```
civ_nutri <- civ_nutri %>% mutate(z_score = (height - medheight) / stdheight) %>% mutate(stunted = ife
civ_2_4years <- civ_nutri %>% filter(age >= 2*12 & age < 5*12)
```

Figure 1: Distribution of z-scores by year

```
ggplot(civ_nutri, aes(x = z_score, y = factor(year))) +
  geom_density_ridges(alpha=0.3) +
  geom_vline(xintercept = -2, linetype = "dashed") +
  scale_fill_discrete() +
  theme_ridges() +
  labs(fill = NULL) +
  theme(legend.position = c(0.98, 0.98), legend.justification = c(1, 1)) + xlim(-6, 6) + xlab("Z-Score
```

2.3.a

```
model_stunted_national <- lm(stunted ~ factor(year)-1, data = civ_2_4years, weights = pweight)
model_stunted_north <- lm(stunted ~ factor(year)-1, data = civ_2_4years %>% filter(north == 1), weight
model_stunted_south <- lm(stunted ~ factor(year)-1, data = civ_2_4years %>% filter(north == 0), weight

# 1. Run the interacted model to get the differences by year
model_diff_ns <- lm(
  stunted ~ factor(year) + factor(year):north - 1,
  data = civ_2_4years,
  weights = pweight
)

# 2. Extract the coefficient matrix
ct_diff_ns <- coeftest(model_diff_ns)

# 3. Isolate ONLY the interaction rows (the North-South differences)
# These will be the coefficients named "factor(year)1988:north", etc.
diff_rows <- grep(":north", rownames(ct_diff_ns))
ct_diff_only <- ct_diff_ns[diff_rows, , drop = FALSE]

# 4. Disguise the row names so huxtable aligns them with the other models
rownames(ct_diff_only) <- c("factor(year)1988", "factor(year)1993", "factor(year)2008")

# 5. Restore the coeftest class
class(ct_diff_only) <- "coeftest"
```

```r
civ_2_4years_no_outliers <- civ_2_4years %>% filter(z_score >= -6 & z_score <= 6)

# Re-estimate the models without outliers
model_stunted_national_no_outliers <- lm(stunted ~ factor(year)-1, data = civ_2_4years_no_outliers, we
model_stunted_north_no_outliers <- lm(stunted ~ factor(year)-1, data = civ_2_4years_no_outliers %>% fi
model_stunted_south_no_outliers <- lm(stunted ~ factor(year)-1, data = civ_2_4years_no_outliers %>% fi

model_diff_ns_no_outliers <- lm(
  stunted ~ factor(year) + factor(year):north,
  data = civ_2_4years_no_outliers,
  weights = pweight
)

# 2. Extract the coefficient matrix
ct_diff_ns_no_outliers <- coeftest(model_diff_ns_no_outliers)

# 3. Isolate ONLY the interaction rows (the North-South differences)
# These will be the coefficients named "factor(year)1988:north", etc.
diff_rows_no_outliers <- grep(":north", rownames(ct_diff_ns_no_outliers))
ct_diff_only_no_outliers <- ct_diff_ns_no_outliers[diff_rows_no_outliers, , drop = FALSE]

# 4. Disguise the row names so huxtable aligns them with the other models
rownames(ct_diff_only_no_outliers) <- c("factor(year)1988", "factor(year)1993", "factor(year)2008")

# 5. Restore the coeftest class
class(ct_diff_only_no_outliers) <- "coeftest"

reg <- huxreg(
  `National`             = model_stunted_national,
  `North`                = model_stunted_north,
  `South`                = model_stunted_south,
  `Diff\n(North-South)`  = ct_diff_only,
  `National `            = model_stunted_national_no_outliers,
  `North `               = model_stunted_north_no_outliers,
  `South `               = model_stunted_south_no_outliers,
  `Diff\n(North-South) `  = ct_diff_only_no_outliers,
   coefs = c(
    "Year 1988" = "factor(year)1988",
    "Year 1993" = "factor(year)1993",
    "Year 2008" = "factor(year)2008"
  ),
  error_format = "({std.error})",
  error_pos = "below",
  align = "c",
  statistics = c(
    N = "nobs",
    R2 = "r.squared"
  ),
  stars = c(`***` = 0.01, `**` = 0.05, `*` = 0.1),
  note = "Significance: {stars}"
)

# 2. Insert the Yes/No row
```

```r
# We insert after row 7 (1 header row + 6 coefficient/error rows)
reg <- insert_row(reg,
  c("Outliers", "Yes", "Yes", "Yes", "Yes", "No", "No", "No", "No"),
  after = 7
)

# 3. Format the new Yes/No row (center the Yes/No text)
reg <- set_align(reg, row = 8, col = 2:9, value = "center")

# 4. Apply your LaTeX stability settings (wrap header & set width)
reg <- set_wrap(reg, row = 1, col = everywhere, value = TRUE)
reg <- set_width(reg, 1)
reg <- set_label(reg, "tbl-stunted_north_south")
reg <- set_caption(reg, "National, North, and South stunting rates for 2-4 years old children excludin
reg <- set_all_padding(reg, 1.5)
reg
```

Figure 2: Distribution of z-scores for 2-4 years old children by year and region

```r
ggplot(civ_2_4years, aes(x = z_score, y = factor(year), fill = factor(north))) +
  geom_density_ridges(alpha=0.3) +
  geom_vline(xintercept = -2, linetype = "dashed") +
  scale_fill_discrete(labels = c("South", "North")) +
  theme_ridges() +
  labs(fill = NULL) +
  theme(legend.position = c(0.98, 0.98), legend.justification = c(1, 1)) + xlim(-6, 6) + xlab("Z-Score
```

2.3.b

```r
model_stunted_sexe_1 <- lm(stunted ~ factor(year)-1, data = civ_2_4years %>% filter(sexe == 1), weight
model_stunted_sexe_0 <- lm(stunted ~ factor(year)-1, data = civ_2_4years %>% filter(sexe == 0), weight

# 1. Run the interacted model to get the differences by year
model_diff_ns <- lm(stunted ~ factor(year) + factor(year):sexe - 1, data = civ_2_4years, weights = pwe

# 2. Extract the coefficient matrix
ct_diff_ns <- coeftest(model_diff_ns)

diff_rows <- grep(":sexe", rownames(ct_diff_ns))
ct_diff_only <- ct_diff_ns[diff_rows, , drop = FALSE]

# 4. Disguise the row names so huxtable aligns them with the other models
rownames(ct_diff_only) <- c("factor(year)1988", "factor(year)1993", "factor(year)2008")

# 5. Restore the coeftest class
class(ct_diff_only) <- "coeftest"

# Models without outliers
model_stunted_sexe_1_no_outliers <- lm(stunted ~ factor(year)-1, data = civ_2_4years_no_outliers %>% f
model_stunted_sexe_0_no_outliers <- lm(stunted ~ factor(year)-1, data = civ_2_4years_no_outliers %>% f
```

```r
# 1. Run the interacted model to get the differences by year
model_diff_ns_no_outliers <- lm(stunted ~ factor(year) + factor(year):sexe - 1, data = civ_2_4years_no

# 2. Extract the coefficient matrix
ct_diff_ns_no_outliers <- coeftest(model_diff_ns_no_outliers)

diff_rows_no_outliers <- grep(":sexe", rownames(ct_diff_ns_no_outliers))
ct_diff_only_no_outliers <- ct_diff_ns_no_outliers[diff_rows_no_outliers, , drop = FALSE]

# 4. Disguise the row names so huxtable aligns them with the other models
rownames(ct_diff_only_no_outliers) <- c("factor(year)1988", "factor(year)1993", "factor(year)2008")

# 5. Restore the coeftest class
class(ct_diff_only_no_outliers) <- "coeftest"

reg <- huxreg(
  `National`             = model_stunted_national,
  `Male`                 = model_stunted_sexe_1,
  `Female`               = model_stunted_sexe_0,
  `Diff\n(Male-Female)`  = ct_diff_only,
  `National `            = model_stunted_national_no_outliers,
  `Male `                = model_stunted_sexe_1_no_outliers,
  `Female `              = model_stunted_sexe_0_no_outliers,
  `Diff\n(Male-Female) ` = ct_diff_only_no_outliers,
   coefs = c(
    "Year 1988" = "factor(year)1988",
    "Year 1993" = "factor(year)1993",
    "Year 2008" = "factor(year)2008"
  ),
  error_format = "({std.error})",
  error_pos = "below",
  align = "c",
  statistics = c(
    N = "nobs",
    R2 = "r.squared"
  ),
  stars = c(`***` = 0.01, `**` = 0.05, `*` = 0.1),
  note = "Significance: {stars}"
)

# 2. Insert the Yes/No row
# We insert after row 7 (1 header row + 6 coefficient/error rows)
reg <- insert_row(reg,
  c("Outliers", "Yes", "Yes", "Yes", "Yes", "No", "No", "No", "No"),
  after = 7
)

# 3. Format the new Yes/No row (center the Yes/No text)
reg <- set_align(reg, row = 8, col = 2:9, value = "center")

# 4. Apply your LaTeX stability settings (wrap header & set width)
# Format the table
reg <- set_align(reg, row = 8, col = 2:9, value = "center")
```

```
reg <- set_wrap(reg, row = 1, col = everywhere, value = TRUE)
reg <- set_width(reg, 1)
reg <- set_label(reg, "tbl-stunted_sexe")
reg <- set_caption(reg, "National, Male, and Female stunting rates for 2-4 years old children.")
reg <- set_all_padding(reg, 1.5)
reg
```

Figure 3: Distribution of z-scores for 2-4 years old children by year and region

```
ggplot(civ_2_4years, aes(x = z_score, y = factor(year), fill = factor(sexe))) +
  geom_density_ridges(alpha=0.3) +
  geom_vline(xintercept = -2, linetype = "dashed") +
  scale_fill_discrete(labels = c("Female", "Male")) +
  theme_ridges() +
  labs(fill = NULL) +
  theme(legend.position = c(0.98, 0.98), legend.justification = c(1, 1)) + xlim(-6, 6) + xlab("Z-Score
```

2.4

```
# 1. Center the log consumption variable and create income tertiles
civ_2_4years <- civ_2_4years %>%
  mutate(
    # Center against the national mean
    log_cons_centered = log(conspc88) - mean(log(conspc88), na.rm = TRUE),
    # Divide into 3 equal-sized groups: 1 = Low, 2 = Medium, 3 = High
    income_tertile = ntile(log(conspc88), 3)
  )

# 2. Estimate the general national model
model_general <- lm(
  stunted ~ log_cons_centered * factor(year) + agehead + femalehead + educationhead,
  data = civ_2_4years,
  weights = pweight
)

# 3. Estimate the stratified models for each income tertile
model_low <- lm(
  stunted ~ log_cons_centered * factor(year) + agehead + femalehead + educationhead,
  data = civ_2_4years %>% filter(income_tertile == 1),
  weights = pweight
)

model_med <- lm(
  stunted ~ log_cons_centered * factor(year) + agehead + femalehead + educationhead,
  data = civ_2_4years %>% filter(income_tertile == 2),
  weights = pweight
)

model_high <- lm(
  stunted ~ log_cons_centered * factor(year) + agehead + femalehead + educationhead,
```

Figure 4: Log household consumption ventiles vs stunting rate for 2-4 years old children by year

```r
civ_2_4years <- civ_2_4years %>% filter(z_score >= -6 & z_score <= 6)

# 1. Create the binned dataset
binned_data <- civ_2_4years %>%
  # Create a log consumption variable if you haven't already
  mutate(log_cons = log(conspc88)) %>%
  # Group by year to ensure we calculate quantiles within each year's distribution
  group_by(year) %>%
  # Chop the log(consumption) data into 20 equal-sized buckets (ventiles)
  mutate(cons_bin = ntile(log_cons, 20)) %>%
  # Group by both year and the new bins to calculate the bucket averages
  group_by(year, cons_bin) %>%
  summarise(
    # Calculate the weighted mean for the x-axis and y-axis
    mean_log_cons = weighted.mean(log_cons, w = pweight, na.rm = TRUE),
    stunting_rate = weighted.mean(stunted, w = pweight, na.rm = TRUE),
    .groups = "drop"
  )

# 2. Plot the binned data
ggplot(binned_data, aes(x = mean_log_cons, y = stunting_rate, color = factor(year))) +
  # Add the binned averages as dots
  geom_point(size = 3, alpha = 0.8) +
  # Overlay the linear regression lines with 95% confidence intervals
  geom_smooth(method = "lm", se = TRUE, level = 0.95, size = 1.2, alpha = 0.2) +
  # Formatting to make it look academic
  theme_minimal() +
    theme(legend.position = c(0.98, 0.98),
      legend.justification = c("right", "top")) +
  labs(
    x = "Log Household Consumption per Capita",
    y = "Stunting Rate (Probability of being Stunted)",
    color = "Survey Year",
    title = "Gradient of Stunting Over Time"
  ) +
  scale_y_continuous(labels = scales::percent_format())
```

```
  data = civ_2_4years %>% filter(income_tertile == 3),
  weights = pweight
)

# 4. Compile the output into a single huxtable
outreg_quantiles <- huxreg(
  `General (All)` = model_general,
  `Low Cons (T1)` = model_low,
  `Medium Cons (T2)` = model_med,
  `High Cons (T3)` = model_high,
  coefs = c(
    "Average Stunting Rate (1988)" = "(Intercept)",
    "Log Consumption (Centered)" = "log_cons_centered",
    "Year 1993 Shift" = "factor(year)1993",
    "Year 2008 Shift" = "factor(year)2008",
    "Cons. Effect Diff (1993)" = "log_cons_centered:factor(year)1993",
    "Cons. Effect Diff (2008)" = "log_cons_centered:factor(year)2008"
  ),
  error_format = "({std.error})",
  error_pos = "below",
  align = "c",
  statistics = c(N = "nobs", R2 = "r.squared"),
  stars = c(`***` = 0.01, `**` = 0.05, `*` = 0.1),
  note = "Significance: {stars}"
)

# Apply formatting
outreg_quantiles <- set_all_padding(outreg_quantiles, 1.5)
outreg_quantiles <- set_label(outreg_quantiles, "tbl-stunting_by_income")
outreg_quantiles <- set_caption(outreg_quantiles, "Heterogeneous Effects of Consumption on Stunting Ra

outreg_quantiles <- insert_row(outreg_quantiles,
  "Controls", "Yes", "Yes", "Yes", "Yes",
  after = 13
)

outreg_quantiles <- insert_row(outreg_quantiles,
  "Outliers Excluded", "Yes", "Yes", "Yes", "Yes",
  after = 14
)

outreg_quantiles
```

Exercise 3

Figure 5: Stock of foreign population (non-citizens) over time

```r
# # validate countries have population and migration data for all years
# migration_pop %>%
#   filter(year <= 2023 & year >= 2013) %>%
#   group_by(country) %>%
#   summarise(
#     n_years = n_distinct(year),
#     n_population = sum(!is.na(population)),
#     n_migration = sum(!is.na(migration_flow))
#   )

# read data/population.csv and data/migration.csv
population <- read_csv(here::here("measurement_p2/data/population.csv"))
migration <- read_csv(here::here("measurement_p2/data/stocks.csv"))

# left join migration with population on country and year
migration_pop <- migration %>%
  left_join(population, by = c("entity", "year")) %>%
  select(entity, year, population, obs_value) %>%
  rename(
    country = entity,
    migration_flow = obs_value
  ) %>%
  mutate(migration_rate = migration_flow / population) %>% filter(year <= 2023 & year >= 2013)

# keep only countries with n_population = 11 and n_migration = 11
migration_pop <- migration_pop %>%
  filter(year <= 2023 & year >= 2013) %>%
  group_by(country) %>%
  filter(n_distinct(year) == 11 & sum(!is.na(population)) == 11 & sum(!is.na(migration_flow)) == 11) %
  ungroup()

# migration rates for the United States, Germany, France, and Poland
selected_migration_rates <- migration_pop %>%
  filter(country %in% c("United States", "Germany", "France", "Poland")) %>%
  select(country, year, migration_rate)

oecd_yearly_stats <- migration_pop %>%
  group_by(year) %>%
  summarize(
    # Method 1 macroeconomic aggregate rate
    world_migration_flow = sum(migration_flow, na.rm = TRUE),
    world_population = sum(population, na.rm = TRUE),
    mean_rate = world_migration_flow / world_population,

    # Weighted standard error calculation for the confidence interval
    sum_pop = sum(population, na.rm = TRUE),
    sum_pop_sq = sum(population^2, na.rm = TRUE),
    weighted_var = sum(population * (migration_rate - mean_rate)^2, na.rm = TRUE) / (sum_pop - (sum_po
    se = sqrt(weighted_var / n()),

    ci_lower = pmax(0, mean_rate - 1.96 * se), # pmax(0) ensures CI doesn't drop below 0%
    ci_upper = mean_rate + 1.96 * se
  )
```